

Course Outline: How to Orchestrate Data

Title: How to Orchestrate Data **Skill:** __ — Skill x: Model debugging and integration **Learnpack:** + ¿Cómo orquestar datos? **File:** s__-w18d53-como-orquestar-datos **Prerequisite:** Background processing (cronjobs, triggers), data pipelines, ETL concepts **Target Audience:** Beginner developers who need to keep their data pipelines running automatically and reliably **Total Lessons:** 8 **Format:** Mixed (12.5% hands-on)

Course Description

A pipeline that runs once is a script. A pipeline that runs reliably, on schedule, and monitors its own health is an orchestrated system. This course explains what data orchestration means, how it differs from a plain cronjob or script, and how to build and monitor a Python pipeline that runs each step in sequence, checks its own output, and reports status. Students finish by writing a small orchestration script that demonstrates the core loop: run → validate → log → repeat.

Course Philosophy

This course emphasizes:

- Understanding WHY orchestration exists before learning HOW to implement it
 - Building monitoring and logging habits from the start — not as afterthoughts
 - Pragmatic orchestration: a well-structured Python script is often sufficient before reaching for heavy tools
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Course Statistics Summary

Section	Lessons
00 - Welcome	1
01 - Data Orchestration	2
02 - Monitoring Pipelines	3
03 - Assessment	1
04 - Outro	1
Total	8

00.0 Keeping Your Data Pipeline Running

Learning Objectives:

- Explain the gap between a one-time script and a production-grade pipeline

- Describe what orchestration adds to a data pipeline

Content Outline:

- The problem: a script runs manually once and produces data — but what happens next week?
- What makes a pipeline "production-grade": scheduling, error handling, logging, monitoring
- Orchestration as the answer: coordinating when things run, in what order, and how failures are handled
- Overview of what this course covers

Transitions: This lesson frames the need. Section 01 explains what orchestration is and what tools exist.

Key Principles:

- A pipeline without monitoring is a blind pipeline — you don't know if it ran or failed
- Orchestration is the practice of making pipelines reliable and observable over time

Examples:

- A daily ETL job that extracts sales data, transforms it, and loads it to a data warehouse — but it silently fails every Tuesday when the source API is slow
 - With orchestration: the job is scheduled, retried on failure, and sends an alert if it hasn't completed by 7am
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01.0 Data Orchestration

Learning Objectives:

- Define data orchestration and explain what it adds beyond a plain cronjob or script
- Describe the key concepts of a DAG-based orchestrator: tasks, dependencies, DAGs
- Preview what lesson 01.1 covers (orchestration tools in practice)

Content Outline:

- **What orchestration means:** coordinating the sequence, timing, and dependencies of pipeline steps
- **Beyond cronjobs:** cronjobs run scripts on a schedule; orchestrators also track state, handle failures, retry, log, and alert
- **DAG (Directed Acyclic Graph):** a way to model tasks and their dependencies — task A must complete before task B starts
 - "Directed" = there's a direction of flow
 - "Acyclic" = no circular dependencies
- **Key orchestration concepts:** task, dependency, run, success/failure, retry, alert
- **When plain Python or cronjobs are enough:** for simple, low-stakes pipelines with 1-3 steps

Transitions: Now you know what orchestration is. Lesson 01.1 looks at real orchestration tools and their core patterns.

Key Principles:

- Orchestration is not a technology — it's a discipline. You can orchestrate with a simple Python script or with Airflow
- The right tool depends on pipeline complexity: simple pipelines don't need Airflow

Examples:

- A DAG: Extract (task 1) → Validate (task 2) → Transform (task 3) → Load (task 4)
 - Task 2 cannot start until task 1 succeeds; if task 1 fails, tasks 2–4 are skipped
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01.1 Orchestration in Practice

Learning Objectives:

- Compare plain Python scripts, cronjobs, and dedicated orchestration tools as pipeline coordination mechanisms
- Identify when each approach is appropriate
- Describe how Airflow and Prefect work conceptually (DAG definition, task execution, UI)

Content Outline:

- **Plain Python script:** simplest orchestration — tasks are functions called in sequence; logging is manual; no retries
 - Use when: 1-3 steps, runs daily or less, error tolerance is high
- **Cronjob:** adds scheduling; script still runs sequentially but on a timer
 - Use when: the pipeline is stable and failures are acceptable; no dependency tracking needed
- **Airflow (conceptual):** DAG defined in Python; Airflow scheduler runs tasks on workers; UI shows run history, failures, retries
 - Strengths: battle-tested, rich UI, extensive integrations
 - Weaknesses: heavy to set up; overkill for small teams
- **Prefect (conceptual):** Python-native orchestration; tasks and flows are Python functions with decorators; cloud or self-hosted
 - Strengths: lightweight, developer-friendly, easy to add to existing Python code
 - Use when: team is Python-first and doesn't need Airflow's ecosystem
- **Practical starting point:** begin with a well-structured Python script, add logging and error handling; only add a dedicated orchestrator when the pipeline complexity demands it

Transitions: You know the tools. Section 02 focuses on monitoring — knowing whether your pipeline ran, whether it produced good output, and what to do when things go wrong.

Key Principles:

- The best orchestration tool is the one your team will actually use and maintain
- Complexity in your orchestration system adds maintenance burden — start simple

Examples: Orchestration decision tree:

- 1-2 steps, monthly run → plain Python script
- 1-3 steps, daily run → cronjob

- 3+ steps, daily run, need retry + alerting → Prefect
 - Enterprise, many teams, complex dependencies → Airflow
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02.0 Monitoring Pipelines

Learning Objectives:

- Explain what pipeline monitoring means and why it matters
- List the key signals to monitor: execution time, failure rate, data freshness, SLA compliance
- Preview the metrics and hands-on content in lessons 02.1 and 02.2

Content Outline:

- **What monitoring means:** tracking pipeline health over time — not just whether it ran, but whether it produced useful output
- **Why monitoring matters:** silent failures are the worst kind — the pipeline appears to run but the output is wrong or stale
- **Key signals to monitor:**
 - **Execution time:** did it complete in the expected window?
 - **Data freshness:** was output produced recently enough?
 - **Error rate:** what fraction of runs failed this week?
 - **Output quality:** does the output pass basic validation (row count, nulls, value ranges)?
- **SLA (Service Level Agreement):** a commitment about when the pipeline will produce output — e.g., "sales data available by 7am every day"
- Overview of 02.1 (health metrics) and 02.2 (hands-on monitoring script)

Transitions: This overview establishes WHY you monitor. Lesson 02.1 goes deeper into which metrics to track and how to define thresholds.

Key Principles:

- A pipeline that runs but produces stale or wrong data is worse than a pipeline that fails loudly — at least a loud failure tells you something is wrong
- Monitoring is part of the pipeline, not an afterthought

Examples:

- A sales pipeline runs at midnight but takes 4 hours. Data is available at 4am. The SLA is 7am. Everything is fine.
 - The same pipeline now takes 6 hours. Data arrives at 6am. Still fine. Next week: 9 hours. SLA is missed. Alerts fire.
 - Without monitoring: no one notices until an analyst reports that today's dashboard is blank.
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02.1 Pipeline Health Metrics

Learning Objectives:

- Define and calculate four key pipeline health metrics: run duration, success rate, data freshness, and output row count
- Set meaningful thresholds for each metric
- Explain the difference between alerting on individual failures vs patterns

Content Outline:

- **Run duration:** how long does the pipeline take? Establish a baseline; alert when it exceeds 2× baseline
 - Track as: `end_time - start_time` per run
- **Success rate:** what percentage of scheduled runs completed successfully?
 - Track weekly: if success rate drops below 80%, investigate root cause
- **Data freshness:** when was the output last updated? The pipeline may have run, but the output might not have changed
 - Critical for downstream consumers: analysts, dashboards, models
- **Output row count:** does the output have the expected number of rows? A sudden drop from 10,000 rows to 50 rows is a signal
 - Set min/max bounds from historical data
- **Alerting strategy:**
 - Individual failure: log it, retry once, then alert if the retry also fails
 - Pattern: if 3 consecutive runs fail → page someone
 - Avoid alert fatigue: don't alert on every transient failure

Transitions: With metrics defined, lesson 02.2 builds a Python script that implements the run → validate → log → status cycle.

Key Principles:

- A metric without a threshold is just noise — always define what "good" and "bad" look like before you start collecting
- Alert on patterns, not individual events — one failed run is noise; three consecutive failures is a problem

Anti-patterns:

- **Logging success only:** failures must also be logged with enough context to diagnose the root cause
- **Alerting on every failure:** this produces alert fatigue — engineers stop responding

Examples: Pipeline health dashboard snapshot:

Metric	This Week	Last Week	Threshold
Success rate	86%	100%	> 80%
Avg duration	47 min	43 min	< 90 min
Freshness (hours since last update)	2.1h	1.8h	< 6h
Avg output rows	9,847	10,123	8,000–12,000

02.2 Building a Pipeline Monitor 🖥️

Challenge Prompt: Write a Python script that orchestrates three pipeline steps in sequence (extract, transform, load), validates each step's output count, and prints a final status summary showing which steps passed and which failed.

Solution Code:

```
import random

def extract():
    print("Extracting data...")
    rows = random.randint(90, 110)
    return rows

def transform(rows):
    print(f"Transforming {rows} rows...")
    transformed = int(rows * 0.95)
    return transformed

def load(rows):
    print(f"Loading {rows} rows...")
    return rows

def validate(step, rows, min_rows=80):
    passed = rows >= min_rows
    status = "PASS" if passed else "FAIL"
    print(f" [{status}] {step}: {rows} rows (min: {min_rows})")
    return passed

results = {}

raw = extract()
results["extract"] = validate("extract", raw)

transformed = transform(raw)
results["transform"] = validate("transform", transformed)

loaded = load(transformed)
results["load"] = validate("load", loaded)

print("\n--- Pipeline Status Summary ---")
for step, passed in results.items():
    print(f"{step}: {'OK' if passed else 'FAILED'}")
print(f"Overall: {'SUCCESS' if all(results.values()) else 'FAILURE'}")
```

Challenge Code:

```

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def transform(rows):
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def load(rows):
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    return rows

def validate(step, rows, min_rows=80):
    passed = rows >= min_rows
    status = "PASS" if passed else "FAIL"
    print(f" [{status}] {step}: {rows} rows (min: {min_rows})")
    return passed

results = {}

# Step 1: run extract and validate its output
raw = extract()
results["extract"] = # your code here

# Step 2: run transform and validate its output
transformed = transform(raw)
results["transform"] = # your code here

# Step 3: run load and validate its output
loaded = load(transformed)
results["load"] = # your code here

# Step 4: print the summary
print("\n--- Pipeline Status Summary ---")
for step, passed in results.items():
    print(f"{step}: {'OK' if passed else 'FAILED'}")
print(f"Overall: {'SUCCESS' if all(results.values()) else 'FAILURE'}")

```

03.0 Data Orchestration Assessment 🧠

Learning Objectives:

- Demonstrate understanding of orchestration concepts and when to use them
- Correctly interpret pipeline monitoring metrics
- Identify alert fatigue anti-patterns and propose better alternatives

Question Format: Multiple choice

Questions:

1. What is the key difference between a cronjob and a dedicated orchestrator like Airflow or Prefect?
 - a) Cronjobs run on Linux; Airflow runs on any OS
 - b) An orchestrator tracks task state, handles retries, manages dependencies, and provides visibility; a cronjob just runs a script on a schedule
 - c) Cronjobs are faster; orchestrators are slower but more reliable
 - d) Orchestrators replace cronjobs entirely; cronjobs should never be used
2. What does "data freshness" measure in pipeline monitoring?
 - a) How many rows the pipeline produced today
 - b) How recently the output was updated — whether consumers are getting current data
 - c) Whether the pipeline ran without errors
 - d) How fast the pipeline completed relative to its SLA
3. A pipeline has run 10 times this week. 3 runs failed. What is the success rate?
 - a) 30%
 - b) 70%
 - c) 75%
 - d) 80%
4. Your pipeline normally produces 10,000 rows. Today it produced 55 rows. What should happen?
 - a) Nothing — small fluctuations are normal
 - b) Log a warning but continue — row count is not a critical metric
 - c) Trigger an alert — a drop from 10,000 to 55 rows indicates a likely data quality or extraction failure
 - d) Increase the minimum row threshold to 100 and re-run
5. A team receives alerts for every single pipeline failure. Engineers start ignoring the alerts. What is this called and how should it be fixed?
 - a) Alert overload — fix by removing all alerts
 - b) Alert fatigue — fix by alerting on patterns (e.g., 3 consecutive failures) instead of individual failures
 - c) Monitoring failure — fix by adding more alerts
 - d) SLA breach — fix by extending the SLA window
6. When is a plain Python script with logging a sufficient orchestration solution?
 - a) Never — always use Airflow for production pipelines
 - b) Only if the pipeline runs once and never again
 - c) When the pipeline has 1-3 steps, runs infrequently, and failures are acceptable or easy to catch manually

- d) Only if the team doesn't know Airflow

7. What does a DAG (Directed Acyclic Graph) represent in data orchestration?

- a) A type of database optimized for pipelines
- b) A graph of tasks and their dependencies, where tasks flow in one direction with no circular loops
- c) A monitoring dashboard that shows pipeline run history
- d) A Python library for scheduling jobs

Evaluation Criteria:

- 7/7: Excellent — strong grasp of orchestration and monitoring concepts
- 5–6/7: Good — minor gaps in metric interpretation or tool selection
- 3–4/7: Needs review — revisit Sections 01 and 02
- Below 3/7: Restart recommended

Score Interpretation: Questions map to: orchestrator vs cronjob (Q1), data freshness (Q2), success rate calculation (Q3), row count alerting (Q4), alert fatigue (Q5), when Python is enough (Q6), what a DAG is (Q7).

04.0 Your Pipeline Never Sleeps

Content Outline:

- Course recap: what students accomplished
 - Understood what orchestration adds beyond plain scripts and cronjobs
 - Learned key orchestration concepts: tasks, dependencies, DAGs
 - Compared Python scripts, cronjobs, Airflow, and Prefect
 - Defined pipeline health metrics: success rate, duration, freshness, row count
 - Set alerting thresholds and avoided alert fatigue anti-patterns
 - Built a pipeline monitoring script that runs, validates, and reports status
- Connection to bigger picture: every ML model depends on fresh, high-quality data arriving reliably — orchestration is what makes that possible
- Next steps: the next modules introduce CLIs and MCPs, which provide another layer of tooling for giving AI agents access to data and capabilities
- Resources for further exploration:
 - Apache Airflow documentation: airflow.apache.org
 - Prefect documentation: docs.prefect.io
 - "Designing Machine Learning Systems" (Chip Huyen) — data pipeline chapters
 - Great Expectations: open-source data validation framework
- Encouragement: a pipeline that runs reliably and alerts you when something goes wrong is engineering at its best — it frees you to focus on what matters

Note: This is conclusion only — no theory, exercises, or assessment.
